

2.1	1	2	3	4	5	6	7	8	9	10	11	12
	G	Q	T	K	B	M	P	A	R	N	I	W
	13	14	15	16	17	18	19	20	21	22	23	
	D	F	V	S	E	U	H	O	L	C	J	
2.2	a. I need your big boat						b. I need that bear					
	c. I need this big ash						d. I need my own meat					
	e. I need that cap over there											
2.3	a. ñi ʈpilaʈɪfɪk'ekiɖ / xekiɖ						b. ñi ñzaqoxekiɖ					
	c. ñi ahaqxekiɖ						d. ñi ʈəjɪkxekiɖ					
	e. ñi ñzrovsk'ekiɖ / xekiɖ											

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Commentary

(From NACLO solution – presumably written by the author.)

How to solve:

Easy step 1

- See that ñi = I
- See that xekiɖ = need (x ɛk' aft er f and s)
- alternate analysis zyæfk' and liysk' end in k' (xɛɛ/aft er k')
- These analyses might not pop out at fi rst, but they are seeable.

Easy step 2

- Count the nouns in the English translations.
- See that fi sh shows up 3 times.
- See that cap shows up 5 times.
- See that hut shows up 1 time.
- See that all other nouns show up twice.

Step 3a

- Look for correspondences in the Nivkh data (looking for fish, cap, and hut)

Step 3b

- Notice 'pila' (could initially mean 'big' or 'over there' by factoring out all of the nouns – cap is tricky, but the others are easy)
- Noti ce ɬo 3 ti mes (after segmentati on of pila), ɬo = fish
- Noti ce ñivra = my hut, so ivra = hut, ñ means my, which is nice because it goes with ñi=I
- Noti ce (h)aq = hut (5 ti mes) – the h will be tricky

Step 4

- figure out if pila is 'big' or 'over there'
- compare ɬpilazyæfk' and tɛpilaɬo = ??(that or this)- pila –fi sh
- Notice that the two words containing pila have diff erent prefixes (t vs. ɬ)
- Look at the other sentences that contain 'big' or 'over there' in their translations.
- zyæfk' = either boat or bear
- Compare I need that boat over there vs I need that fish over there
- Notice that the prefixes don't match
- Compare I need this big fish with I need your big bear
- Notice they have different prefixes, so tɛ = this, ɬ=your
- Now you can match up 13 with D, 19 with H, 7 with P and 8 with A, ...

Step 5

- You still have ɬo and a -ɬo – one means a fish and the other means that fish over there
- Notice a -mu and ñ-mu. ñmu is my _____. Of the nouns that appear twice, boat, branch, and picture have 'my' in their translation.

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- Only 'boat' also goes with 'that over there', none have 'a' translations. So mu = boat, ñ = my, and a = 'that over there'.
- So 1=G, 16=S, 14=F and 20=O

Step 6 : Cap

- aq shows up 5 times (it is haq, and x ʔʔ/after a prefix) :
 - 2: ñi hæ-haq-xekiɖ
 - 5: ñi ñ-aq-xekiɖ
 - 12: ñi ʔ-aq-xekiɖ
 - 17: ñi ñ-it'ulv-haq-xekiɖ
 - 21: ñi haq-xekiɖ
- Meanings :
 - B : 'I need my cap'
 - E: 'I need my winter cap'
 - L; 'I need a cap'
 - Q: 'I need that cap'
 - W: 'I need your cap'
- evident that 12=W, 5=B, 17=E (so it'ulv(h) = winter, but the 'h' shows up in 21 too, so it'ulv)
- a cap vs that cap, we know 'a' = no prefix on fi sh, and haqxekiɖ is obviously less prefixed than hæhaqxekiɖ, so haq = cap (h deletes)
- after a C, or after 'my' and 'your') hæ = that
- 21=L, 2=Q

Step 7

- hæ-zaqo and ʔ-zaqo = that X and your X, so zaqo = knife and 9=R and 18=U
- liysk' and ʔ-liysk' = a X and your X, so liysk' = wolf and 10=N and 23=J
- ʔ-efl aŋk and ñ-efl aŋk = your X and my X, so 22=C and 4=K
- p'-ezŋai and ʔ-ezŋai = my own X (we haven't seen p' before, but we know ʔ is 'your') and your X, so 6=M and 15=V
- i-təfk' and təfk' = his/her X (by elimination) and a X, so 3=T and 11=I

The only thing left was the vowels of the prefixes of a few forms. I took the offenders out. I modified the answers so that there are 2 right answers for 2 of the Eng to Nivkh translations.